# Privacy as Articulation Work in HIV Health Services

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### ABSTRACT
Normative accounts on health information privacy often highlight the importance of regulating data sharing. Yet, little attention has been paid to how health professionals perform and negotiate privacy practices in highly multidisciplinary, technologically-mediated medical work. This paper examines information privacy practices in two HIV outpatient clinics based in two NHS hospitals in London (UK). Methods include 46 semi-structured interviews, primarily with health professionals and technology developers, ethnographic observation and document analysis. Drawing on an empirically informed understanding of privacy as 'articulation work', we focus on the indeterminate nature of information privacy practices and examine the work required to translate privacy, from a normative professional duty to an enacted medical practice. This analysis also highlights the invisibility of privacy practices and their coordinating role in delivering technologically-supported medical care. The paper ends with a discussion of implications for practice and technology design.

### Author Keywords
Information privacy; medical confidentiality; articulation work; HIV; design.

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### INTRODUCTION
The importance of health information sharing and interoperability among healthcare providers has been underlined in a number of policy initiatives in different countries [12, 25]. Yet, striking a balance between information accessibility and patient confidentiality in technologically mediated healthcare work remains a challenge. Privacy considerations have long accompanied plans for the implementation of electronic records and integrated systems in healthcare [28]. In the UK, 186 serious data breaches were reported to the Department of Health between 2011-2012 [13] and efforts to alter the technological landscape in the English National Health Service (NHS) raised privacy and confidentiality concerns in recent years [2, 10, 31].

Against this background, patient confidentiality remains a highly valued professional duty and a cornerstone of medical practice [20]. Legal provisions and codes of ethics outline standards for protecting patient information, provide a framework for appropriate professional conduct and specify repercussions from a confidentiality breach. Although useful as templates for professional practice, these normative accounts do not fully capture the lived experiences of clinicians who manage patient confidentiality within health settings. Healthcare provision often involves unexpected events and information needs that are not always predictable. In addition, medical work is becoming increasingly multidisciplinary, including not just a wider range of health professionals, but also social care and other occupations.

To better support service delivery and technology use, it is important to gain an empirical understanding of how information privacy and confidentiality are enacted in practice to manage the collaborative and distributed nature of medical work. Recent literature reviews in the area emphasise the importance of empirical research that transcends the individual level of analysis, to use case studies and ethnographic methods looking at collaborative, rather than individual, privacy practices from the perspective of data users [4, 5, 11, 23, 27, 34, 35, 38].

This paper focuses specifically on collaborative information privacy practices from the perspective of health professionals working in outpatient clinics for people living with the Human Immunodeficiency Virus (HIV) in two NHS hospitals in London (UK). HIV health services provide a unique context to study privacy and confidentiality practices because of the sensitivity of HIV patient information and the potential for stigmatisation. To


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account for confidentiality concerns, HIV health services in the UK were historically organised in isolation to mainstream care, using segregated practices for service delivery and patient management. With current treatments extending life expectancy, however, HIV is increasingly managed in ways similar to other chronic conditions that require strong collaboration and information exchange between different medical specialties [1]. As medical care extends beyond a single discipline or health setting, HIV patient information is also often necessary beyond the boundaries of HIV clinics to support appropriate medical decisions and safe care. This paper examines how health professionals manage patient confidentiality using a number of different systems at the interface between HIV and mainstream care. The analysis informs the debate on shifting privacy practices for HIV care and contributes to the discussion on how these would be best technologically supported.

The rest of this paper is structured as follows. The next section provides the background literature review, followed by the theoretical framework underpinning this work. The research methods are then presented, together with a section on HIV health settings explaining how HIV patient care is organised in the UK. Key findings on privacy as articulation work and as an invisible practice are contained within the next two sections. The discussion follows, including implications for design and practice, and the paper ends by highlighting the main conclusions from this work.

## LITERATURE REVIEW

A large body of literature has been devoted to examining privacy as a set of elements, states or categories. Westin [43] observes that there are four basic privacy states between an individual and society: solitude, intimacy, anonymity and reserve. Similarly, Gavison [15] argues that secrecy, anonymity and solitude constitute the three elements of privacy, importantly highlighting, however, that these are: 'distinct and independent, but interrelated, and the complex concept of privacy is richer than any definition centred around only one of them'. The richness and complexity this conveys permeates many contemporary views.

Instead of looking for a common denominator to what constitutes privacy, Solove [36] focuses on conceptualising privacy problems and proposes a taxonomy of information harms. More recent work discusses the relevance of context relevant informational norms in framing an understanding of privacy as contextual integrity [24]. Following a similar approach, Palen and Dourish [26] conceptualise interpersonal privacy as dynamically and dialectically managed in response to competing tensions and circumstances in sociotechnical environments.

In this work, we adopt a view of information privacy as a set of contextual and situated practices, drawing on recent work that seeks to overcome the rigidity of previous privacy conceptualisations and to problematise the fluidity of public/private distinctions [17, 21, 22, 24, 32, 36]. We are interested in privacy not as an abstract trade-off, a technical property or a set of preferences, but as an inherent part of interrelated and performative social and cultural practice [14]. This approach transcends the idea of privacy as fixed and stable, to draw attention to its ongoing collaborative accomplishment.

Although significant work has been done on collaborative practices in health settings from a CSCW perspective, privacy has more often been examined as an individual value or preference, or as a consequence, rather than as a collaborative practice in its own right [5, 35]. Only recently have researchers focused more explicitly on how privacy is managed and enacted at the group-level [11, 23].

One of the studies within this strand of research specifically examined the different mechanisms and workarounds emergency care staff used to enact privacy while carrying out medical work [23]. Drawing on the theory of contextual integrity, another ethnographic study in three different settings outlined three types of group dynamics and discussed the implications for appropriate privacy management [11]. Both studies point to the challenges of aligning privacy policies and technological design within highly distributed, collaborative work practices in dynamic environments.

The work presented here focuses on a different context, that of HIV health services, to expand the body of literature looking at privacy as a highly situated collaborative practice. In doing that, we employ concepts of articulation and invisible work to examine how privacy practices are accomplished in practice, as explained in the following section. This introduces a novel theoretical conceptualisation for studying privacy, which holds potential for illuminating aspects more difficult to clearly articulate as technical requirements or specifications, or to stipulate in organisational policies.

## THEORETICAL FRAMEWORK

According to Gerson and Star [16], articulation work is defined as a series of coordination tasks carried out to support work practices and respond to local contingencies within highly complex collaborative settings. This is achieved 'by packaging a compromise that 'gets the job done', that is closes the system locally and temporarily so that work can go on' [16]. In this respect, articulation work refers not only to the activities and tools employed to cope with the distributed nature of cooperative work, but also denotes mechanisms of interaction and communication, which require continuous articulation work in themselves to successfully fulfill their role [29].

As a theoretical concept, articulation work sensitises towards the 'continuous efforts required in order to bring together discontinuous elements […] into working configurations' [40]. This brings to the fore elements of background practices that are often not clearly included in formal representations of work, but remain crucial for the accomplishment of collaborative tasks [3, 16, 29, 39]. The idea of 'invisible work' also relates closely to articulation as it seeks to highlight informal tasks performed in the background with discretion [37].

A number of CSCW studies have encompassed articulation work as a crucial aspect in making sense of and addressing the implications of distributed, collaborative work across different settings [3, 19, 39, 42]. This has been particularly highlighted in studies looking at the management and use of patient records in health settings [6-8]. Yet, this theoretical lens has not been explicitly considered in understanding how privacy and confidentiality practices are produced and configured. This paper argues that privacy work contains significant elements of articulation in the process of translating a normative professional duty to an enacted medical practice, work that often happens silently in the background. An emphasis on articulation allows invisible practices that are 'relegated to a background of expectation' [37] to come to the fore and become explicitly recognised [40]. This has implications on who gets to define what counts as privacy work, how it should be done and how it should be effectively designed in record keeping systems.

The case studies presented here highlight how privacy practices are embedded in managing the distributed nature of medical work within HIV centres and across hospital departments. In this, we look not just at how privacy practices are part of the articulation work necessary to carry out clinical tasks, but also at how privacy practices themselves require articulation work to be accomplished. This highlights how privacy becomes socially and collaboratively produced, in ways that are not always formally acknowledged in organisational policies and codes of conduct, or supported by technological systems.

## METHODS

This paper draws on two case studies on privacy and confidentiality practices in HIV clinics. Fieldwork began in October 2009 with a pilot project in the HIV unit of a teaching hospital in the South East of England. Following that, the main part of the research was carried out in two HIV outpatient specialist centres in east London between April and October 2010. These sites were in the process of negotiating changes to the way they managed HIV patient information, from using stand-alone systems to integrating their HIV records with the rest of the hospital. These negotiations opened a productive space for engaging in privacy discussions with staff members and brought to the fore different experiences with providing integrated HIV care while protecting patient information.

A total of 46 semi-structured interviews were conducted, primarily with health professionals, including HIV medical consultants and other specialties, junior doctors and registrars, nurses and nursing assistants, health advisors, administrative employees, IT developers, a pharmacist and a patient representative. The majority of staff looking after HIV patients in the clinics and most of the IT staff involved in the integration negotiations were interviewed, following purposive and snowballing sampling methods. Relevant 'experts' from academia, government agencies and community organisations who had an insight on the integration project were also involved in interviews to provide some background and contextualise privacy practices for HIV care. Table 1 presents a breakdown of participants based on their roles. Each interview lasted one hour on average. Interview questions evolved throughout the study and were adapted for different professional groups, but high-level topics included perceptions around privacy and confidentiality practices in HIV care, experiences with using paper and electronic records, challenges emerging from the use of segregated technological systems and practices adopted to overcome communication and collaboration barriers.

| Roles | # |
|---|---|
| Consultants | 10 |
| Registrars and junior doctors | 3 |
| Nurses & nursing assistants | 10 |
| Health advisors | 5 |
| Pharmacist | 1 |
| IT developers | 7 |
| Administrative staff | 4 |
| General management | 2 |
| 'Experts' | 3 |
| Patient representative | 1 |

**Table 1. Roles of participants.**

Interviews were supplemented by ethnographically informed observation in the two HIV outpatient centres for an average of 2-3 days a week over a period of six months. Areas observed included outpatient waiting rooms, reception areas and different staff offices (HIV nurses, doctors and health advisors). The researcher also observed a number of meetings looking at HIV clinical care, multidisciplinary meetings discussing patient cases, service provision, training as well as executive and management meetings, among others.

Document analysis was also used to inform the interpretation of findings. Minutes of meetings between 2005-2010, hospital policies, risk registers, risk assessment forms, confidentiality policies, clinic guidelines, incident reports, codes of conduct, HIV process flow diagrams, discussion papers and other documents related to the integration negotiations were reviewed and analysed where relevant to guide fieldwork and contribute to the findings.

The data from the interviews, observation and documentation were analysed using thematic analysis with emerging codes iteratively added to a pre-defined coding framework [44]. Ethical approval for the project was granted by University College London Hospital (UCLH) Research Ethics Committee Alpha (ref. 09/H0715/78).

## SETTING: HIV HEALTH SERVICES

According to the most recent prevalence statistics produced by the Health Protection Agency (HPA), an estimated 100,000 people live with HIV in the UK, out of whom one fifth have not been diagnosed [18]. By the end of 2012, the number of patients accessing HIV care had doubled since 2003, with the total number reaching 77,610 individuals. The majority of cases and new diagnoses are reported in London, with a higher prevalence in the most deprived boroughs of the east where this research was focused.

In the UK most HIV specialist outpatient centres operate models of anonymous testing and treatment, due to the increased level of confidentiality restrictions applicable to sexual health information, as set out in legislation and professional codes of practices. Although medical confidentiality remains important in all medical settings, HIV clinics have historically adopted particular privacy practices: they use their own unique identifiers and are not obliged to link patients' identities to their NHS numbers as routinely done in other settings; they also use separate paper and electronic records which cannot be shared with health professionals even when co-located in the same hospital.

The two HIV centres studied here had adopted these practices, with record-keeping systems and information flows historically designed to retain the segregation of patient information. The HIV units were thus using their own stand-alone clinical support systems, which did not communicate with the hospital Electronic Patient Record (EPR). This entailed separate confidentiality practices and distinct privacy regimes between the HIV clinics and mainstream hospitals, which raised interesting questions as to how privacy practices would be managed in collaborative medical work at the intersection of HIV and general patient care.

This way of organising HIV services, however, is being questioned as HIV has shifted from being a terminal condition to one where chronic treatment is available [1, 30, 33, 41]. The long-term nature of the disease and the changing demographics of patient populations create new challenges as to how heterogeneous and distributed practices of HIV care will be best technologically supported. This led the British HIV Association (BHIVA) to recommend integrated clinical records for the management of HIV patients, therefore encouraging a shift from HIV-specific to generic confidentiality practices as relevant for any other patient in a hospital setting [9]. Adoption of this recommendation had been slow but provided interesting opportunities for examining privacy practices as the two HIV outpatient centres were still operating at the interface of mainstream care and HIV models at the time this research was carried out.

## HIV PATIENT TRAJECTORIES

In the two clinics, HIV patients were seen either for follow-up routine appointments every 3-4 months or on a walk-in basis. Two weeks before their scheduled appointment they visited the clinic to have blood taken, usually by a nurse. They then attended on the day to receive the results, discuss with the doctor and, if necessary, receive drug prescriptions.

Drop-in clinics served HIV patients with semi-acute problems between regular appointments. Apart from face-to-face patient clinics, HIV professionals frequently held what they called virtual clinics, where they used paper and electronic records to discuss patients with their team.

To account for the medical and social implications of an HIV diagnosis, care was multidisciplinary and team-based within the HIV units. Psychologists and social workers were often involved to support patients. Also, occupational therapists offered treatment and rehabilitation services to people living with HIV who encountered problems with everyday tasks as a result of their condition or who needed help after being discharged from the hospital. Other services were also involved on a needs basis, ranging from dieticians to mental health specialists. To protect from stigma and accidental disclosure within the clinics and become embedded with the larger patient body, people living with HIV were not referred to as HIV patients but as 'medical' patients in the HIV clinics.

The care of pregnant women living with HIV was transferred to the hospitals' antenatal departments, where a team of HIV specialists and obstetricians followed them until delivery. By adhering to therapy, receiving specialist obstetric care and avoiding breastfeeding, they could significantly reduce the chances of vertical (mother-to-child) transmission. After the pregnancy, care continued normally in the HIV clinics.

When the HIV specialist unit was closed, patients presented to the Accident & Emergency (A&E) Department or other NHS walk-in clinics. One of the hospitals had no dedicated inpatient ward for patients whose problems were primarily HIV-related so they were transferred to a different site. There were still patients living with HIV, however, who were hospitalised for problems not directly associated to their positive status. These patients were under the care of different specialties acting on a liaison basis with HIV doctors on matters of drug interactions and other complications. Apart from such ad hoc communication, few connections existed between HIV and other medical teams. In the second hospital, the HIV team coordinated care in an inpatient ward where most people living with the condition were admitted, although this was not a formally dedicated HIV ward.

## ARTICULATION WORK & PRIVACY PRACTICES

The segregation of record-keeping systems between HIV clinics and mainstream hospital departments entailed increased articulation work (in the sense of background practices) to manage collaboration needs in multidisciplinary teams inadequately supported by existing information infrastructures. Due to the condition often considered stigmatising, articulation work was done to fill in the gaps created by the privacy practices adopted in the two hospitals and to sustain absolute confidentiality for HIV patients.

### Creating HIV patient trajectories

When a patient was admitted to hospital, they were either triaged in emergency admissions or sent to the acute care unit directly through a referral from their General Practitioner (GP), before being admitted to other hospital wards if necessary. Patients living with HIV often required the attention of an HIV clinician as part of their multidisciplinary team. With little information sharing between the HIV centres and mainstream hospital departments, articulation of HIV patient trajectories often took place in ad-hoc and contingent ways, as the junior doctor below explains:

*How we get informed […] varies. It doesn't always work completely. Most patients who are admitted to the hospital come through the Acute Care Unit and the HIV Liaison Nurse was going to their ward round every morning. […] And also the hospital teams, there's information that basically if they have any patient with HIV they should inform us. And sometimes it comes through the Pharmacy if they are written up for ARVs [HIV medication] and we get informed. So […] any patients who have HIV tests in the hospital, any positive results get phoned through to either our Health Advisors or [the HIV Liaison Nurse] again.* (Junior Doctor)

The role of the liaison nurse was to provide the link between acute admissions and HIV clinics, carrying out the background work that allowed the initiation of HIV patient trajectories. This was necessary in order to be able to draw on HIV-specific expertise, but also to gain access to the information contained in the separate set of patient notes held at the HIV centre for patients already treated there (as the notes were not allowed to leave the HIV centre). Articulation work was done to attach the appropriate medical team and draw a relevant care plan, thereby overcoming the implications of distributed work and retaining HIV-specific confidentiality.

### Working with HIV patient records

For HIV patients who attended scheduled appointments in the hospital, another layer of articulation work had to be performed. In one of the hospitals studied, the antenatal clinic for pregnant women living with HIV took place in the obstetrics department rather than in the HIV centre. This meant that attending HIV Consultants had to bring a number of resources together to support clinical interactions: paper notes had to be transferred from the HIV centre, access to one of the HIV-specific systems needed to be set-up in the obstetrics department, while also drawing on the mainstream hospital system. Often in collaboration with a midwife and an obstetrician, HIV Consultants coordinated across these resources, to acquire the relevant background to the consultation and document medical encounters in a way that would support further multidisciplinary care, both in obstetrics and crucially also in the subsequent trajectory of the patient back in the HIV centre after labour:

*Now there, I'm in the antenatal clinic and there I manage all the patients on EPR. […] When I'm in [the HIV centre] I use [System 1] and [System 2] as my prime sources and then if I have to look up EPR it will be a subsidiary thing. There [antenatal clinic] they are on EPR. Their blood results are done on EPR, everything is on EPR and I use [System 1] as a subsidiary thing. But I have [System 2] in both places and I put the data in from both into that. Of course you've got to bear in mind at all times I will have the paper record as well. And in the Antenatal Clinic I will have the paper record from the hospital, I will have the paper record from [the HIV centre], I will have the woman's handheld notes and in terms of previous pregnancies I'll have all her previous handheld notes and if I want to find out about the baby I will have another set of paper notes as well. So, and that's as well as the computer. (HIV Consultant)*

Articulation work for the care of pregnant women living with HIV was necessary because the infrastructure developed to support the obstetrics department did not take in account HIV specificities neither did it allow communication and information exchange with the HIV centre. Since the main hospital system provided few options in supporting the HIV privacy culture, this meant clinicians had to use a number of different practices to address this difference, engaging in articulation work to accomplish privacy in practice: articulating what medical confidentiality meant for people living with HIV, while also supporting the tasks necessary to retain this sense of confidentiality in collaborative practices within HIV centres and across boundaries.

Crossing the boundaries of different privacy regimes also involved physical movement to a different space when HIV technologies and patients were not co-located:

*[…] then you maybe have to see the patient in one room and then go back to another room to write their prescription and check their results. (Junior Doctor)*

Similar information seeking, translation and coordination work to support collaborative practice was carried out in many departments within the same hospital, since HIV patients could have different types, versions and combinations of electronic records and paper notes. Often health professionals had to use resources additional to their electronic and paper records to obtain a more complete picture of a patient's health and social situation, by phoning colleagues, establishing contact with other departments, holding meetings or following-up with liaison or community nurses. Of interest here is the articulation work performed to allow people to work well together across privacy regimes, while still keeping their record-keeping practices separate, driven by the need to respond to privacy concerns about perceptions of stigmatisation around HIV.

**Engaging in discovery and translation**

In the clinics, an investigation and discovery process similar to the one followed in cross-boundary collaboration with other settings, was also performed. To produce a view of the patient that informed and guided locally and temporally relevant clinical decisions, HIV professionals drew on a number of sources within the centres: one system containing patient demographic information, another system containing prescriptions and two more systems primarily including blood results, as well as HIV paper notes - each with different or overlapping parts of patient health information. Multiple systems were used to provide the required functionality in the clinics, but also to allow for privacy options in terms of different access levels between staff. Depending on their role, some would only have access to the system containing demographic information, while others would have access to all systems, having to coordinate the information to be able to provide appropriate care while still protecting patient confidentiality.

The HIV nurse quoted below, provides an example of how she would read the information fragments included within each of the different systems, comparing and contrasting between them to support HIV care:

*You look through [System 2] to check the trend of their viral loads, CD4 and if they are for blood tests, which blood tests they need to have and whether they need any other bloods, whether they are immune to hepatitis B or they need a vaccination. Then you check [System 3] as well, to check on a result like their glucose, their NEPs and their full blood count and their liver and kidney to see if they were abnormal last time. And then you check [System 4] for their previous appointment, whether they've been attending and whether they have future appointments because that will determine whether they need more medications or not. Yeah. Whereas [System 2] would have told you when that day was, it's keeping their prescriptions, you know, that you can ask them why did you run out of medications. (HIV Nurse)*

Another consultant discussed this as a type of translation process between the interfaces available through the different systems, where each supported the clinical interaction differently, while sometimes information was incongruent and remained at the discretion of the clinician to make sense of the discrepancies:

*The interfaces are different and the data I can get about the patient in each place are different and the way you get the data in each system is completely different. There are a few bits with common data in them but on the whole all three are holding different things about patients. The demographics ought to match on a good day but then sometimes the name might be spelt slightly differently or there will be something that isn't quite the same. (HIV Consultant)*

Often this process of information discovery and translation of records was not formalised or planned, but depended on the clinicians deciding which aspects of existing record-keeping practices were deemed relevant and important:

*It's just very, kind of dependent on a particular clinician knowing and utilising selective bits of each system to find out the whole thing. (HIV Consultant)*

**Accessing technological systems**

Being able to use the different record-keeping systems in the first place required further background work as each system had to be accessed separately using different log-in details. Although designed as security mechanisms, the large number of passwords and the rate with which they changed, meant that in practice they could compromise confidentiality, as they had to be easily remembered or shared to fulfil clinical work, or could compromise medical care, if they posed obstacles even to legitimate access to patient records. As the following clinician notes in relation to the work required for retrieving her details and gaining access to patient information:

*[…] I try to impose some human values on it. So I will make sure I've got one password for them but they change their rhythm of passwords at different times, so I've always got three passwords, and I lock myself out regularly because I've put the wrong one in to the wrong system […] 30% of the time I have to get help to actually log on because I've put in the wrong thing to start me off. (HIV Consultant)*

This section discussed how privacy-driven background work was embedded into producing HIV patient trajectories, records and clinical interactions, especially when it came to more intricate and complicated tasks. This work was made necessary in the first place because of the increasingly collaborative nature of HIV care in the context of information sharing restrictions in the two hospitals studied. In essence, articulation work was done to manage the privacy implications of providing multidisciplinary care to patients living with HIV, both within outpatient centres and across the boundaries with other hospital departments.

In turn, this articulation work was also an integral component of sustaining privacy practices specific to HIV care.

## INVISIBILITY OF PRIVACY PRACTICES

During fieldwork, health professionals explained how translating the professional ideal of confidentiality in practice was not always straightforward, but was done in silent and invisible ways. Many privacy practices were successful exactly because they were part of background articulation work; they remained largely invisible to patients and often to health professionals outside HIV clinics, yet done in way that communicated the information relevant to accomplish clinical tasks.

### Record-keeping

Staff members in the HIV clinics had developed their specific codes to fulfil HIV confidentiality requirements in their record-keeping practices. For example, to be able to easily distinguish paper notes belonging to HIV patients against general sexual health service users, clinics used coloured stickers attached to the front of the notes. Marked HIV paper records where unintelligible to outsiders but still easy to identify by staff members, therefore avoiding confusion between notes when retrieving and classifying folders, while also providing the potential to protect information with higher levels of sensitivity more efficiently by locking them in separate cabinets. Staff members also commented on how they tried to reverse paper records so that the front page containing the patient's name would not be visible, although this seemed much more difficult to achieve in practice.

### Clinical consultations

Privacy practices were also invariably performed in clinical interactions with individual patients in ways subject to the discretion of HIV professionals. When clinicians were in the consultation rooms, they communicated with the reception of the HIV centre through an electronic patient list on their computer screen. As soon as a patient attended for their appointment, their name would change colour to inform the clinician that they were waiting to be called in the room. The appointment list, however, included the names of all patients who would be seeing the same clinician on the day. This meant that staff members had to make sure the list was not visible on the screen when the patient entered the room or that it did not appear as they were switching between the different systems to look at patient information:

*Then you bring [the patient] in and you sit them down, but they can see all these names [pointing to the patient list on the computer screen]. So it's very important to click on their name only. (Nursing Assistant)*

### Interacting with other departments

HIV care had to be done even more invisibly in the main hospitals to protect patient confidentiality. This did not only entail bringing together segregated records and aligning medical practices, but also involved how clinicians physically moved and behaved to manage the privacy implications of their recognisability. Many doctors discussed how they turned their identity badges over so that HIV patients on the hospital wards would not feel that they were being stigmatised or to avoid accidental disclosure to relatives or healthcare staff who would be able to see their specialty. One of the consultants described HIV doctors as '*a walking breach of confidentiality*' and explained how they have to be '*alert to what [they] are embodying in what [they] do*'.

Interviewees also referred to the invisibility of privacy practices when discussing how electronic records were discreetly attributed the role of informing health professionals about a patient's HIV status. Due to the fragmented and distributed nature of patient records, patients did not always fully understand what parts of their information were shared and how between clinicians. As one A&E Consultant suggested, many HIV patients presenting to A&E expected the system to inform healthcare staff about their status automatically, especially when it came to treating open wounds and potential transmission risk. In such an open-plan, busy environment patients could find but few opportunities for disclosing sensitive information such as their status. Therefore, technology was attributed the role of discreetly, invisibly articulating sensitive information and essentially protecting patient dignity and confidentiality:

*You know the system should have made sure that the staff knew so that they as a patient don't have to worry about always having. They don't like having to announce their status in a, 'I'm a leper kind of way'. They would much rather that the system discreetly informs the staff and that their consultation appeared to be unaffected by this bug. […] To have to sit there going, 'You need to be careful because I'm infectious', is not a pleasant place. It reinforces a stigma that they would prefer was handled more discreetly. (A&E Consultant)*

Another example of privacy practices where information protection was done invisibly relates to blood tests and drug prescriptions. Instead of printing patient names and other identifiable information on the label attached to the blood vials, as was the norm for other hospital departments, HIV clinics used their own unique identifiers. This meant that test results would then return to the HIV clinic through a separate path, rather than be entered on the main hospital system. Drug prescriptions were also printed on paper and delivered to the pharmacy by the patients themselves, instead of being sent electronically as for the rest of hospital patients.

Although this was intended to protect patient confidentiality, it also meant that HIV medical work was less recognised in the rest of the hospital. HIV professionals often narrated cases where paper referrals to other departments were rejected because they had not been sent electronically from the main hospital system. This meant that HIV professionals had to explain their practices and justify the invisibility of their privacy work to other clinicians:

*Having the X-ray department sort of ringing us up and saying: 'Oh we can't take pieces of paper, only GPs send us bits of paper.' We said: 'No, no we do as well.' (HIV Consultant)*

'Doing' privacy meant that HIV medical work became less visible and subsequently less legitimised when other hospital practices had been streamlined electronically and homogenised through the integration of records on the hospital system. For privacy practices to reinforce HIV-specific confidentiality, however, they needed to be invisible, in the sense of being performed in the background with the agreement of all parties involved. When this agreement broke down, the legitimacy of these practices was questioned. This difference required articulation work to sustain the privacy afforded to HIV patients and to enforce recognition of the circumstances of HIV care.

## DISCUSSION

This paper looks at how health professionals working in a collaborative and technologically-mediated context describe the accomplishment of privacy practices. To understand the nuances in their narratives we have employed articulation work as a novel theoretical approach to study privacy. Although articulation work has often been examined in studies of medical work [8, 39, 42], little attention has been paid to how it can elucidate aspects of collaborative privacy practices in healthcare. Privacy studies have not often engaged in empirically driven analysis, while previous work has also tended to view privacy as a consequence or implication, rather than a social and collaborative practice.

This work focuses on HIV health services where multiple confidentiality practices are employed in patient care. With HIV management changing as a result of the long-term nature of the condition, HIV professionals are called to manage the ongoing tensions between treatment and side-effects, co-morbidities and an ageing patient population, medication adherence, drug interactions and resistance, the potential for transmission in HIV pregnancies and late diagnoses. As a result information needs extend beyond the boundaries of HIV clinics, with health professionals in other specialties increasingly required to provide care for HIV patients.

In the interest of 'reconciling incommensurate assumptions and procedures' [16] embedded in historically segregated information practices, health professionals engage in background work to accomplish medical encounters, both within HIV centres and across hospital departments. The articulation work depicted in their narratives does not just relate to the achievement of clinical tasks, but also constitutes privacy work in its own right, as it is done to facilitate collaboration across privacy regimes with the aim to sustain patient confidentiality.

Confidentiality in healthcare settings happened invisibly and silently, as for example when HIV professionals discussed how they made sure their appointment list was not on the computer screen when they saw a patient or how they covered their name badges to avoid disclosure by inference. To paraphrase Suchman [40]: 'the better [privacy] work is done, the less visible it is to those who benefit from it'. Attention to articulation work further points to the difficulties HIV professionals faced when they tried to 'construct an arena of voice' [37] that would make their privacy work recognisable across settings (e.g. getting pharmacy prescriptions or radiology referrals accepted in the main hospital) and to maintain a common information space.

By looking at privacy practices as articulation work, one can begin to understand the nuanced and detailed mechanisms health professionals employ in managing multiple and conflicting confidentiality requirements. This expands our understanding of privacy by taking in account the local contingencies [16] entailed in doing privacy in different contexts and generates further insight into how compromise becomes relevant to make things work from a confidentiality perspective. It also supports the view that privacy practices are not episodic and fragmented, but continuous and interrelated. While often presented as clear-cut ethical requirements, translation in practice requires multiple processes of articulation that require careful disentanglement.

### Design implications

Looking at confidentiality practices as articulation work draws attention to the complex ways confidentiality is produced and maintained within record-keeping practices. When composing models of work, it is important to capture the detailed ways through which health professionals translate normative imperatives of confidentiality in practice. Privacy practices done invisibly and silently might be easier to neglect when designing systems to support collaborative work.

In the hospitals studied, existing systems were not able to manage multiple co-existing confidentiality practices as required by HIV specificities (and potentially other conditions). The large number of passwords required and the difficulties logging on to the relevant systems when necessary also created frustrations. As HIV care practices were not adequately supported, health professionals had to engage in increased articulation work to carry out medical tasks while keeping confidentiality in check. This created an ongoing tension between requirements for HIV-specific and general patient confidentiality, which led to negotiations for the integration between the systems used in the HIV clinics and the EPR in the main hospital.

Issues with role-based access control systems and 'all-or-nothing' security controls have also been identified in previous work [e.g. 11]. Although useful in certain situations, these security mechanisms have not yet been developed to the extent that they are able to manage privacy dynamically in response to competing requirements and changing circumstances. A conceptual understanding of privacy as articulation work would allow an in-depth understanding of the mechanisms that facilitate articulation work for confidentiality management while medical tasks are being accomplished.

Instead of introducing rigid mechanisms and homogenising privacy regimes, an alternative would be to work towards understanding and developing information infrastructures with the potential for ambidexterity. This would mean incorporating the capability to handle different degrees of confidentiality (e.g. in the context of HIV health services this would entail HIV-specific and general medical confidentiality). This would accommodate, rather than eradicate, difference for people living with HIV and other conditions, stigmatising or not. Design methodologies should engage with the multiple degrees of confidentiality relevant in medical work for the same patient under different circumstances and acknowledge the articulation work performed as part of and for privacy practices. Privacy enhancing technologies and alternative authentication solutions that embrace the flexibility required in dynamic and collaborative healthcare settings might further support medical work, although their implementation remains slow.

## CONCLUSION

This paper examined how confidentiality practices are performed within the information seeking, translation and coordination work conducted to support HIV health services. As a result of historically segregated practices between HIV centres and main hospital departments, clinicians had to engage in articulation work to coordinate privacy practices across different contexts. Even within clinics, background articulation work was necessary to combine information from multiple systems, which were used to provide the required functionality, but also to allow for privacy options. The analysis highlighted how privacy practices are often done invisibly and in the background of clinical tasks. Many of these privacy practices remained largely invisible to patients and often to health professionals outside HIV clinics, yet were done in way that communicated the information relevant to accomplish clinical tasks.

Differences in record-keeping practices, however, meant that HIV medical work was less visible by other specialties in the main hospital, thus becoming less legitimised. We suggest that design methodologies taking better account of privacy as articulation work could allow the development of patient record systems that concurrently support differing privacy needs in different contexts.

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